

PROJECT DESIGN PHASE-II

Technology Stack (Architecture & Stack)

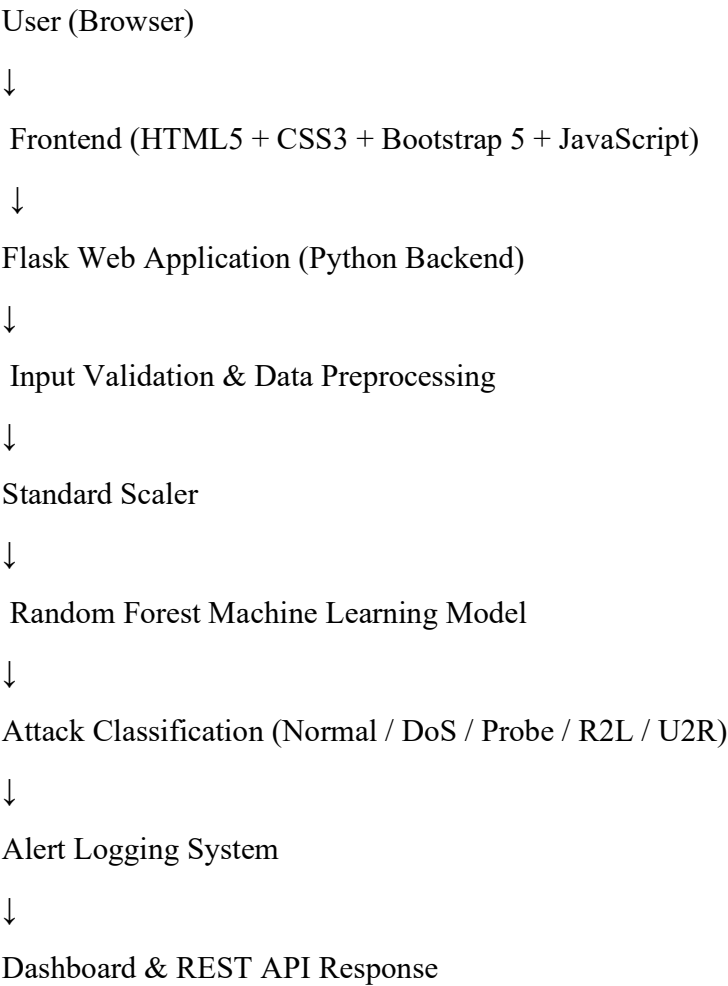
Date: 19 June 2026

Team ID: ----

Project Name: AI-Enhanced Intrusion Detection System

Technical Architecture

Architecture Flow



Infrastructure Demarcation

Local Environment:

- Flask Application Server
- Machine Learning Model (.pkl)
- Alert Logs
- Dataset Storage

Cloud Ready Components:

- REST API Integration
- Cloud Deployment Support
- Remote Monitoring Dashboard

Data Storage Components

- NSL-KDD Dataset
- Pickle Model Files
 - ids_model.pkl
 - scaler.pkl
- Alert Log Storage

External Interfaces

- REST API Endpoint (/api/predict)
- Web Dashboard
- Browser-Based User Interface

Machine Learning Integration

- Random Forest Classifier
- StandardScaler Preprocessing Layer
- Scikit-learn ML Pipeline

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web-based interface for intrusion detection and monitoring	HTML5, CSS3, Bootstrap 5, JavaScript
2	Application Logic-1	User request handling and routing	Python Flask
3	Application Logic-2	Data preprocessing and feature scaling	NumPy, Pandas, StandardScaler

S.No	Component	Description	Technology
4	Application Logic-3	Intrusion detection and classification	Scikit-learn Random Forest Classifier
5	Database	Storage of alerts and prediction logs	Python In-Memory Storage / JSON Logs
6	Cloud Database	Future cloud database integration	MongoDB Atlas / Cloud SQL (Optional)
7	File Storage	Dataset and model storage	Local File System (.csv, .pkl)
8	External API-1	Programmatic intrusion detection service	Flask REST API
9	External API-2	Dashboard analytics integration	Chart.js
10	Machine Learning Model	Detect and classify network intrusions	Random Forest Classifier
11	Infrastructure (Server / Cloud)	Application deployment environment	Local System / Cloud Server (AWS, Azure, GCP)

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks and libraries used in development	Flask, Scikit-learn, NumPy, Pandas, Bootstrap 5
2	Security Implementations	Input validation, API security, secure model access	Flask Security, Input Validation, HTTPS Support
3	Scalable Architecture	Modular architecture separating UI, backend, and ML layers	Three-Tier Architecture
4	Availability	Continuous access through local or cloud deployment	Flask Server, Cloud Hosting
5	Performance	Fast intrusion prediction and response generation	Random Forest Model, Scikit-learn Optimization

Technology Stack Summary

Frontend

- HTML5
- CSS3
- Bootstrap 5
- JavaScript
- Chart.js

Backend

- Python 3
- Flask Framework

Machine Learning

- Scikit-learn
- Random Forest Classifier
- StandardScaler

Data Processing

- Pandas
- NumPy

Dataset

- NSL-KDD Intrusion Detection Dataset

Model Storage

- Pickle (.pkl)

Deployment

- Local Server
- AWS / Azure / GCP (Future Deployment)

Version Control

- Git
- GitHub

Conclusion

The AI-Enhanced Intrusion Detection System utilizes a modern three-tier architecture combining a responsive web interface, a Flask-based backend, and a machine learning inference layer. The

technology stack is lightweight, scalable, and suitable for real-time intrusion detection, cybersecurity demonstrations, academic projects, and future cloud deployment.